

**DASM II LCAM
Installation Manual**

August 1999

PN 39-400078 R01

DASM II LCAM Installation Manual

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*The World Resource
for Precision Signal Technology*

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Revision 01

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Safety Precautions

Cautions Symbols

The following symbols appear in this manual:



CAUTION: Hazardous Voltages present. Reduce the risk of electric shock and danger to personal health by following instructions.



CAUTION: Advises against actions that could damage equipment, produce inaccurate data, or invalidate a procedure.

Mesures de sécurité

Symboles

Vous trouverez ci-dessous la signification des différents symboles utilisés:



ATTENTION: présence de tensions dangereuses. Pour éviter les risques d'électrocution et de danger pour la santé physique, veuillez suivre les instructions.



ATTENTION: risques de blessures corporelles et de dégâts matériels. Veuillez suivre les instructions.

Sicherheitsvorkehrungen

Symbole

Die Symbole in diesem Handbuch haben folgende Bedeutung.



ACHTUNG: Gefährliche Spannungen. Anweisungen befolgen, um Stromschläge und Verletzungen zu vermeiden.



ACHTUNG: weist auf Handlungen hin, die das Gerät beschädigen könnten, unrichtige Daten verursachen oder einen Vorgang auslösen könnten.



CAUTION: SHOCK HAZARD – Due to the risk of electrical shock, this instrument must be installed only by qualified personnel.

ATTENTION: RISQUE DE CHOC ÉLECTRIQUE - Risque de choc électrique. Cet instrument doit être installé SEULEMENT par une personne qualifiée.

ACHTUNG: Elektroschockgefahr - Wegen Elektroschockgefahr darf dieses Instrument nur von Fachleuten installiert werden.



CAUTION: The integrated circuits in the DASM II LCAM are sensitive to electrostatic discharge (ESD). Be sure to follow proper procedures for handling, replacing, storing and transporting ESD-sensitive components.

ATTENTION: Les circuits intégrés du DASM II LCAM sont sensibles aux décharges électrostatiques (ESD). S'assurer de suivre les bonnes méthodes de manutention, de remplacement, d'entreposage et de transport des composants sensibles aux perturbations des décharges électrostatiques (ESD).

ACHTUNG: Die eingebauten Schaltkreise in den DASM II LCAM sind auf Elektrostatik (ESD) empfindlich. Vorschriftsmäßige Handhabung, Auswechslung, Lagerung und Transport von Elektrostatik-empfindlichen Komponenten befolgen.



CAUTION: Be sure the DASM II LCAM is unplugged before attempting to access the fuse.

ATTENTION: Vérifiez que le DASM II LCAM est débranché avant de tenter d'accéder au fusible.

ACHTUNG: Feststellen, daß der DASM II LCAM Stecker herausgezogen ist, bevor an die Sicherungen herangegangen wird.



CAUTION: You must use a fuse of the correct type and rating for the currently configured line voltage.

ATTENTION: Il faut utiliser un fusible de la capacité et du type appropriés pour la tension secteur configurée actuellement.

ACHTUNG: Typ und Belastbarkeit der verwendeten Sicherung müssen für die eingestellte Netzspannung angezeigt sein.

The following general safety precautions must be observed during all phases of service and repair of this equipment. Failure to comply with these precautions or with specific warnings elsewhere in this manual violates safety standards of design, manufacture, and intended use of this equipment.

Do Not Operate In An Explosive Atmosphere

Do not operate the equipment in the presence of flammable gases or fumes. Operation of any electrical equipment in such an environment constitutes a definite safety hazard.

Keep Away From Live Circuits

Operating personnel must not remove equipment covers. Only Qualified Service Personnel may remove equipment covers for internal subassembly or component replacement or any internal adjustment. Do not replace components with power cable connected. To avoid injuries, always disconnect power and discharge circuits before touching them.

Do Not Substitute Parts Or Modify Equipment

Because of the danger of introducing additional hazards, do not install substitute parts or perform any unauthorized modification of the equipment.

Power Cord Connection



CAUTION: DASM II LCAM is designed to work with single-phase power systems having a grounded neutral conductor. To reduce the risk of electric shock, do not plug DASM II LCAM products into any other type of power system. Contact your facilities manager or a qualified electrician if you are not sure what type of power is supplied to your building.

ATTENTION: DASM II LCAM sont conçus pour fonctionner avec des alimentations monophasées munies d'un conducteur neutre mis à la terre. Pour écarter les risques d'électrocution, ne pas brancher de DASM II LCAM un autre type d'alimentation secteur. En cas de doute quant au type d'alimentation électrique du local, veuillez vous adresser au directeur de l'exploitation ou à un électricien qualifié.

ACHTUNG: DASM II LCAM sind für den Betrieb an Einphasen-Stromnetzen mit geerdetem Nulleiter vorgesehen. Um die Stromschlaggefahr zu reduzieren, schieß Sie DASM II LCAM nicht andere Stromquellen an. Ihr Betriebsleiter oder ein qualifizierter Elektriker kann Ihnen die Daten zur Stromversorgung in Ihrem Gebäude geben.

REGULATORY COMPLIANCE

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Shielded cables must be used with this unit to ensure compliance with the Class A FCC limits.

EC Declaration of Conformity

We, Analogic Corporation
Centennial Industrial Park
8 Centennial Drive
Peabody, Ma. 01960

declare under sole responsibility that the: DASM II LCAM Data Acquisition System Manager, to which this declaration relates, meets the essential health and safety requirements and is in conformity with the relevant EC Directives listed below using the relevant section of the following EC standards and other normative documents;

EC EMC Directive 89/336/EEC

Essential health and safety requirements relating to electromagnetic compatibility.

EN 55011:1991 + AM1:1997 and Am2:1996 Group 1, Class A
Limits and methods of measurement of radio disturbance characteristics of information technology equipment.

IEC 801-2:1991 Electrostatic discharge requirements for industrial process, measurement and control equipment.

IEC 801-3:Draft Radiated electromagnetic field requirements for industrial process, measurement and control equipment.

IEC 801-4:1988 Electrically fast transients for industrial process, measurement and control equipment.

IEC 801-5:Draft Testing and measurement techniques - Section 5: Surge Immunity.

EN60601-1-2 Collateral Standard: Electromagnetic compatibility - Requirements and tests.

EC Low Voltage Directive 72/23/EEC

Essential health and safety requirements relating to electrical equipment designed for use with certain voltage limits.

EN 60950 (1992), A1+A2+A3+A4+A11 Safety of information technology equipment including electrical business equipment.

The safety issues of this information technology equipment type have been evaluated by a government accredited European third party organization such as ETL ID No. 0344.

EC Medical Device Directive 93/42/EEC Annex II Section 3.

Essential health and safety requirements Annex (I) relating to electrical medical equipment (Medical Devices). Conformity Assessment by TÜV Rheinland ID No. 0197, Annex II Certificate No. HD 9910023.

The CE mark has been affixed on the device according to Article 17, of the EC Medical Device Directive 93/42/EEC.

Date of Manufacture: Refer to Nameplate

Serial Number: Refer to Nameplate

A handwritten signature in cursive script, appearing to read 'Robert H. French'.

Vice President QA & RA

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Introduction

PURPOSE OF THIS MANUAL

The DASM II LCAM Installation Manual describes how to set up and test the Data Acquisition System Manager (DASM) II Laser Camera interface (LCAM). It includes information on installing hardware, connecting cables, startup, shutdown, troubleshooting, and other related topics.

AUDIENCE

This manual is intended for any person responsible for installing or maintaining the DASM II LCAM.

DOCUMENT STRUCTURE

The DASM II LCAM Installation Manual contains three chapters and four appendices, as follows:

Chapter 1	Presents an Overview of the DASM II LCAM
Chapter 2	Describes installation and start-up including cabling information
Chapter 3	Describes configuring the DASM II LCAM
Appendix A	Describes General Safety Precautions and provides safety specifications
Appendix B	Presents Troubleshooting Information
Appendix C	Presents Technical Diagrams and Functional Descriptions
Appendix D	Lists DASM II LCAM Specifications

Chapter 1

Overview of the DASM II LCAM

1.1 **PRODUCT DESCRIPTION**

The DASM II LCAM enables the user to transfer images from a host computer to a laser camera. The host computer communicates with the DASM II LCAM via the SCSI bus. (See Figure 1-1.)

Circuitry is contained on one printed circuit board assembly. The functionality of this circuitry may be broken into two logical blocks; the Data Acquisition System Manager (DASM) and the Laser Camera interface (LCAM).

The DASM II LCAM is packaged in stand-alone enclosure containing the power supply and all connectors to the SCSI and external devices.



CAUTION: SHOCK HAZARD – Due to the risk of electrical shock, this instrument must be installed only by qualified personnel.

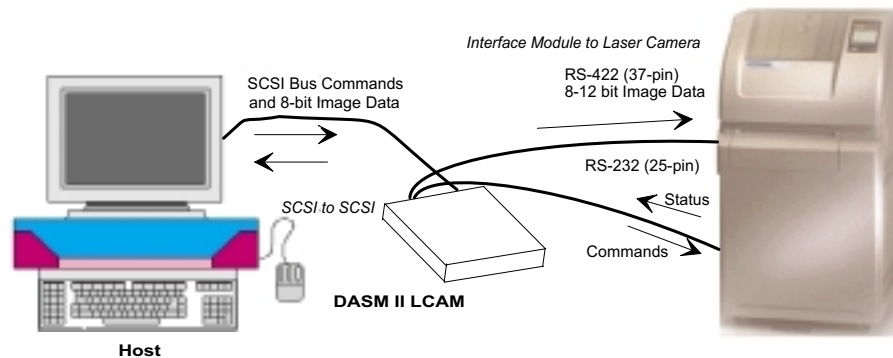
1.2 DASM II LCAM FUNCTIONAL OVERVIEW

The DASM II LCAM performs the following tasks:

- ❑ Accepts digital images from the host, over the SCSI bus
- ❑ Communicates with the laser Camera using the 952 protocol
- ❑ Transfers Images over RS485 bus

The figure below shows the DASM II LCAM in a standard operating environment:

Figure 1-1: DASM II LCAM Functional Flow



Chapter 2

Installation and Start-Up

2.1 OVERVIEW

This chapter explains how to connect the cables and start the DASM II LCAM.

The DASM II LCAM comes in a standard configuration. If you plan to change the configuration, see Chapter 3.

2.2 UNPACKING THE UNIT

The DASM II LCAM consists of the following items:

- ☐ DASM II LCAM
- ☐ SCSI cable
- ☐ Power cord

Note that the customer must provide the following items:

- ☐ Image Cable
- ☐ Serial Camera Control Cable

Verify that you have everything.

2.3 CABLING INFORMATION

The DASM II LCAM connects to the host's SCSI port with a 50-pin SCSI Compliant cable. Two additional shielded cables connect the DASM II LCAM to the laser camera: a camera control serial-line cable, and a 37 conductor image cable.



CAUTION: All units must be powered-down before connecting the interface cables.

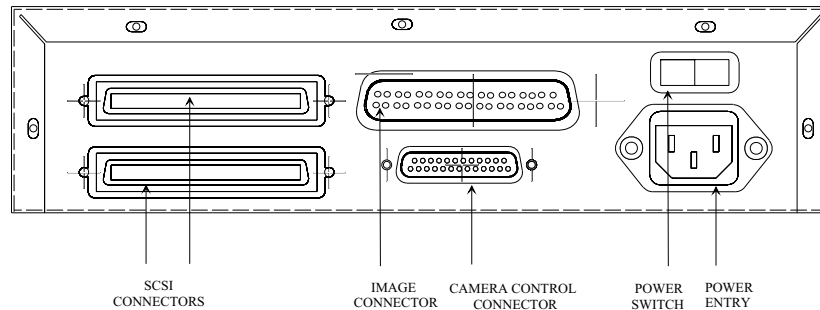
2.3.1 SCSI Connections

The figure below shows the SCSI connectors at the rear of the DASM II LCAM enclosure.



CAUTION: All units must be powered-down before connecting the interface cables.

Figure 2-1: DASM II LCAM Rear Panel



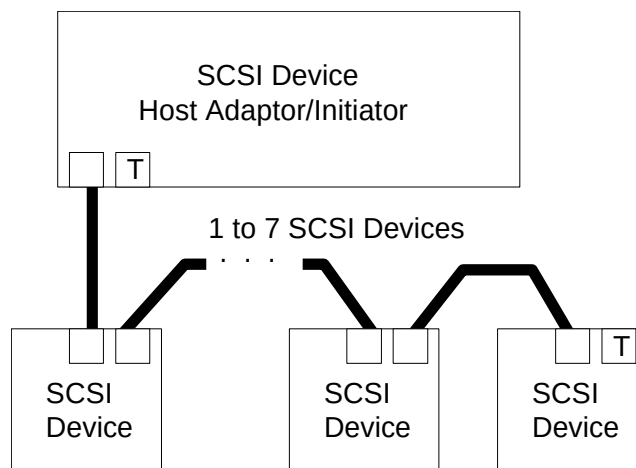
The SCSI cables must be connected, with the strain relief clips locked in place, as follows:

- ❑ The SCSI cable between the host and DASM connects to one of the DASM II LCAM's SCSI ports.
- ❑ The DASM II LCAM's other SCSI port; must be either terminated **OR** connected to the next daisy-chained SCSI device. See the following section, *Connecting Other SCSI Devices*, for more information on daisy-chaining SCSI devices.

2.3.2 Connecting Other SCSI Devices

The DASM II LCAM can be "daisy-chained" with other SCSI devices. Up to 8 SCSI devices (including the host) may be daisy-chained together, as illustrated in the figure below:

Figure 2-2: Daisy-Chaining SCSI Devices



T Indicates SCSI Termination

The SCSI devices at both ends of the cable must be terminated.

- ❑ When DASM II LCAM is not the last device in the chain the SCSI terminator must be removed from the DASM II LCAM's SCSI Connector **AND** the Active SCSI Termination must be turned OFF.
- ❑ Otherwise, when the DASM II LCAM is the last device in the chain, a SCSI terminator must be connected **OR** the Active SCSI Termination Switch must be set ON.

Chapter 3 shows the location of the Active Termination Switch.

Note: The length of all the SCSI cables in a chain must be less than 6.0 meters.

2.3.3 *Laser Camera Control*

The 25-pin 'D' Connector on the rear panel of the DASM II LCAM provides the RS232 or RS422 Serial Line used to control the Laser Camera. Pin assignments are located in Appendix D Specification.

2.3.4 *Image Port*

A 37-pin 'D' Connector on the rear panel rear of the DASM II LCAM provides image data to the laser camera. Pin assignments for this connector are also in Appendix D.

2.4 *START-UP*

This section describes the start-up sequence, as well as how to recognize error conditions.

2.4.1 *Powering On*

This following sequence occurs on the front panel's LEDs when the power switch is turned on:

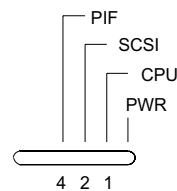
1. The power LED lights.
2. The octal SCSI ID flashes briefly. The Least Significant Bit is next to the power LED. Note that if the SCSI ID is zero, no LEDs flash.
3. These LEDs flash once and go out.

Once the power-up sequence completes, the DASM II LCAM CPU LED (second light from right) blinks regularly, indicating that the CPU is active and functioning normally.

2.4.2 DASM II LCAM LED Indicators

Four LEDs provide a visible indication of the DASM II LCAM's status and error information. These lights are located on the DASM II LCAM front panel as illustrated below:

Figure 2-3: Status Indicators



From left to right, the lights indicate the following:

- ❑ PIF - Personality Interface light indicates a data transfer from the Common Memory to the LCAM circuitry.
- ❑ SCSI - This light indicates the host is accessing the DASM via the SCSI bus, such as commands sent or data received.
- ❑ CPU - This light indicates that the DASM CPU is active and operating properly. It flashes continuously ("blinks" on and off) following power-up.
- ❑ PWR - Power Light indicates that the DASM is powered. If this light does not illuminate at power-up, you should first check the power cord connections.

2.4.3 *Start-Up Sequence*

When the DASM II LCAM is powering up, it performs start-up diagnostics; and other related actions.

At power-up, the DASM II LCAM performs the following tests:

- ☐ EPROM checksum test
- ☐ MFP 68901 access test (the Multi-Function Peripheral chip controls the serial port, timing, and I/O)
- ☐ SCSI register access test
- ☐ Static RAM test
- ☐ Dynamic RAM test
- ☐ DMA test

When the DASM CPU LED blinks continuously at about two flashes per second, the initialization sequence is complete.

If an error occurs, the error condition is reported by the CPU, SCSI, and PIF LEDs blinking in tandem followed by an error code.

Chapter 3

Configuring the DASM II LCAM

3.1 OVERVIEW

This chapter describes how to configure the DASM II LCAM.

3.2 USER SETTINGS

The DASM II LCAM is initially set up as follows:.

1. External Switches:
 - SCSI ID set to one
 - SCSI Termination Switch set to OFF position
 - Serial Camera Control set to RS-232 position
2. Self-Test Options:
 - RAM diagnostic disabled
 - Manufacturing diagnostics disabled
 - Self-Test loop disabled

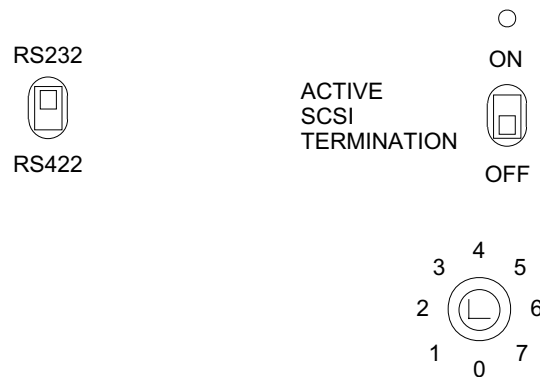
3.2.1 SCSI ID and SCSI Termination

The SCSI ID Switch is accessible from the bottom of the DASM II LCAM enclosure. With a small flat blade screwdriver simply rotate the switch shaft to the desired setting.

Setting the Active SCSI Termination switch to ON enables the internal SCSI termination. If you use this method to terminate, do not install an external terminator in the SCSI port in the rear of the DASM II LCAM enclosure.

If lit, the LED above the Active Termination switch indicates Active SCSI termination is turned ON, and that termination power is being supplied.

Figure 3-1: DASM II LCAM Bottom



3.2.2 Serial Port Setting

The Camera Control serial port can be set for either RS-422 or RS-232 by using the switch at the bottom of the DASM II LCAM enclosure. See Figure 3-1 above.

3.3 REMOVING THE DASM II LCAM COVER



CAUTION: SHOCK HAZARD – Due to the risk of electrical shock, this instrument must be installed only by qualified personnel.

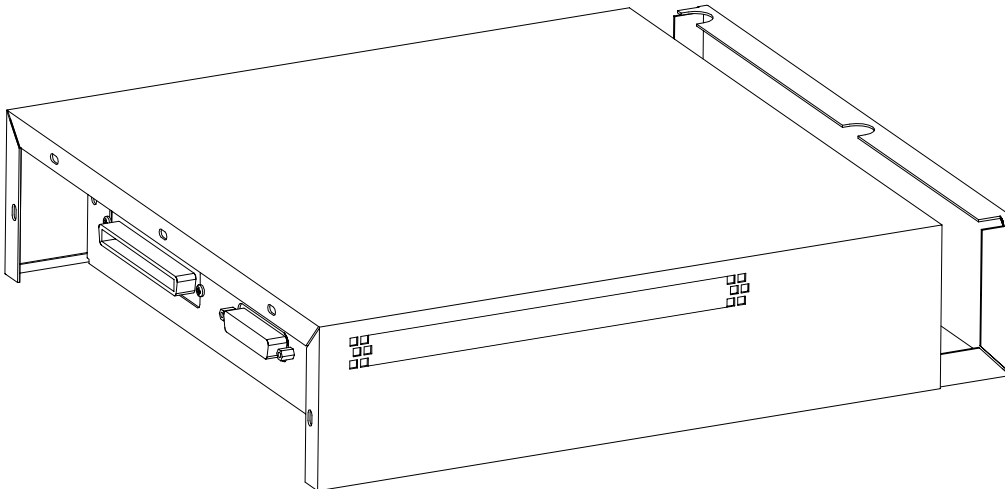
To select Self-Test options it will be necessary to open the DASM II LCAM enclosure and access the printed circuit board. See Figure 3-3 for the location of DIP switch SW1.

To remove the DASM II LCAM enclosure cover:

1. Perform shutdown and power off the DASM.
2. Unplug the power cable from the rear of the DASM.
3. Using the Phillips head screwdriver, remove the five screws on the back of the DASM enclosure. These five screws secure the system enclosure cover. See Figure 3-2.

To remove the enclosure cover, pull it to the rear for approximately $\frac{1}{2}$ to $\frac{3}{4}$ inches, so that the tabs in the cover are free from the chassis bends making up the front panel. Then lift it up and away from the DASM enclosure.

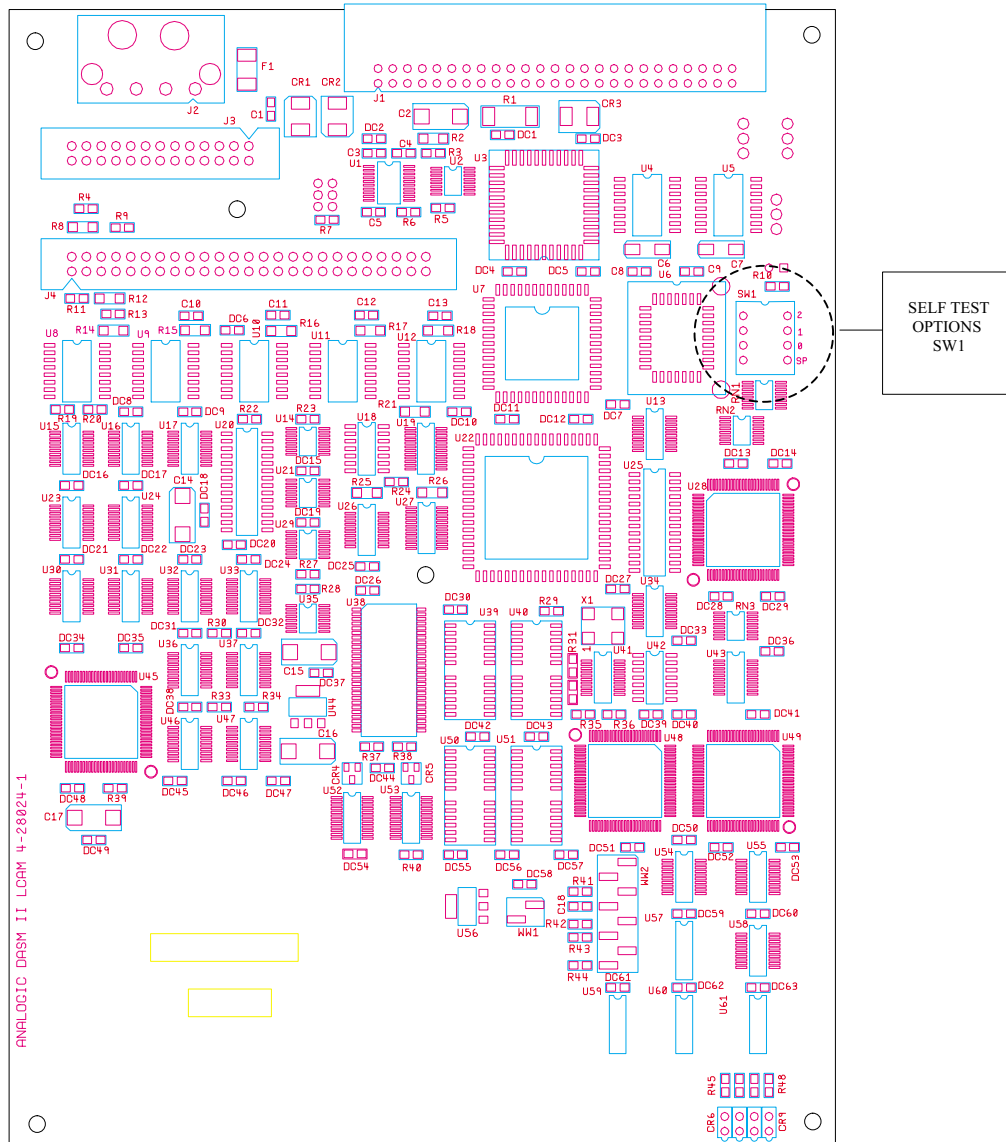
Figure 3-2: DASM II LCAM Enclosure



3.4 DASM II LCAM BOARD

The figure below shows the DASM II LCAM printed circuit board:

Figure 3-3: DASM II LCAM Board



Appendix A

Safety

A.1 INTRODUCTION

This appendix points out standard precautions to be observed while working with the DASM II LCAM.

A.2 EQUIPMENT PRECAUTIONS



CAUTION: SHOCK HAZARD – Due to the risk of electrical shock, this instrument must be installed only by qualified personnel.

Properly power-down the DASM II LCAM and disconnect the line cord from the power entry connector at the rear of the DASM II LCAM before accessing any internal components. Only qualified repairmen should open the DASM II LCAM enclosure.

A.3 POWER REQUIREMENTS

Input range: 100-240VAC
Frequency: 50-60Hz
Current: 0.6A

A.4 POWER SUPPLY

A.4.1 Safety Warning

The power supply manufacturer gives the following warning; “Beware of internal lethal voltages due to charged capacitors even after AC input is disconnected. A minimum of 5 minutes should be allowed after AC Input power is disconnected before handling”.

A.4.2 Power Supply Fuse (Chart)



CAUTION: Be sure the DASM II LCAM is unplugged before attempting to access the fuse.



CAUTION: You must use a fuse of the correct type and rating for the currently configured line voltage.

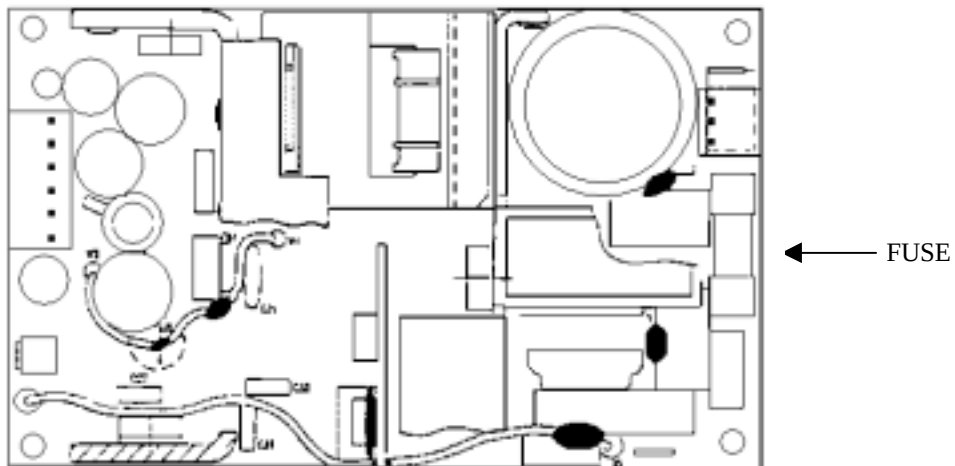
Listed below are the only two approved power supply replacement fuses.

Table A-1: FUSE Specification table

Analogic Part Number	Vendor P/N	Vendor Name
15-500193	21602.5	LITTLEFUSE INC.
15-500193	0001.1008	SCHURTER

A.4.3 Power Supply Layout

The power supply fuse is located as indicated below.



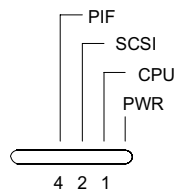
Appendix B

Troubleshooting

B.1 DASM II LCAM LED INDICATORS

The DASM II LCAM front panel contains four LEDs that are used for status and error information. These lights are illustrated below:

Figure B-1: Status Indicators



From left to right, the lights indicate the following:

- ❑ PIF - Personality Interface light indicates a data transfer from the Common Memory to the LCAM circuitry.
- ❑ SCSI - This light indicates the host is accessing the DASM via the SCSI bus, such as commands sent or data received.
- ❑ CPU - This light indicates that the DASM CPU is active and operating properly. It flashes continuously ("blinks" on and off) following power-up.
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B.2 *START-UP SEQUENCE*

When the DASM II LCAM is powering up, it performs start-up diagnostics and other related actions.

At power-up, the DASM Controller performs the following tests:

- ☐ EPROM checksum test
- ☐ MFP 68901 access test (the Multi-Function Peripheral chip control the serial port, timing, and I/O)
- ☐ SCSI register access test
- ☐ Static RAM test
- ☐ Dynamic RAM test
- ☐ DMA test

When the DASM CPU LED blinks continuously at about two flashes per second, the initialization sequence is complete.

If an error occurs, the error condition is reported by blinking the CPU, SCSI, and PIF LEDs in tandem followed by an error code.

B.2.1 *Powering On*

This following sequence occurs on the front panel's LEDs when the power switch is turned on:

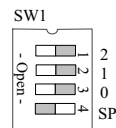
1. The power LED lights.
2. The octal SCSI ID flashes briefly. The Least Significant Bit is next to the power LED. Note that if the SCSI ID is zero, no LEDs flash.
3. These LEDs flash once and go out.

Once the power-up sequence completes, the DASM CPU LED (second light from right) blinks regularly, indicating that the CPU is active and functioning normally.

B.3 SELF-TEST OPTIONS

Once the start-up sequence is complete, the resident software reads the state of the DIP switch labeled SW1 (see Figure 3-3 DASM II LCAM Board) to determine whether further tests are enabled. To access SW1 you must first remove the top cover. Upon locating SW1 you will notice the numbers 0, 1, and 2 silk screened on the printed circuit board (see below).

Figure B-2: LCAM SW1



The way Self-Test executes is determined by the setting shown below:

Switch Position	Single Pass	Continuous Mode	Single Pass W/Memory	Debug Mode
0	Closed	Open	Closed	Open
1	Closed	Closed	Closed	Open
2	Closed	Closed	Open	Open

Note: SW1 position marked "SP" is not used.

Single Pass is the default setting, and is the only setting allowed when connected to a laser camera. The start-up diagnostics execute once, as described in B.2 Start-Up Sequence.

Continuous Mode: After executing the start-up diagnostic, the resident software will continuously run the LCAM Self-Test. Self-Test will continue until the first SCSI command is received by the DASM. Please be aware that when position 0 is open error and other messages encountered during start-up diagnostics are directed to the serial port. Otherwise the errors are encoded on the LEDs.

Single Pass W/Memory is controlled by Bit 2. When open the DASM exercises the entire DRAM during the start-up diagnostics. The Single Pass W/Memory test can take several minutes to complete, this is why it is normally disabled.

Debug Mode: When position 1 is open the program jumps to the I/O Monitor when a diagnostic error occurs. The I/O Monitor gives the basic tools to debug the hardware. Opening positions 0 and 2 allows the technician access to the start-up diagnostic suite via the serial port.

B.4 START-UP PROBLEM

To report diagnostic conditions, the LEDs on the DASM II LCAM front panel have a corresponding binary representation. The figure below shows the value assigned to each LED. The section after the figure explains how the LEDs represent error codes. Note that this section deals with error codes related to the DASM Controller, where Section B.5 explains the method used to report an error from the LCAM diagnostic.

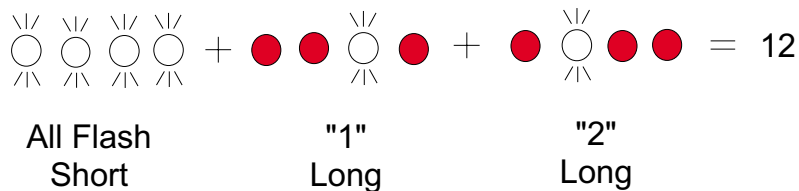
B.4.1 DASM Controller LED Error Codes

Using LEDs 2 and 4, a binary code signals any of the conditions listed in Table B-1 on the following page. The power light and PIF LEDs are not used to report diagnostic conditions.

The three LEDs permit a combined total of only seven error codes. To overcome this limitation, the codes are defined as a series of one or more patterns. Each pattern begins with all LEDs flashing briefly, to indicate the start of the number sequence that follows.

For example, to display error code 12 (Failed SCSI Interface Test), the CPU LED displays the first digit (1), stopping briefly before the SCSI LED displays the second digit (2), as shown in the figure which follows.

Figure B-3: Error Code Indicators



The display of each digit lasts approximately four times the duration of the initial binary "7" (that is, the flash of all LEDs). The pattern then repeats. Only the significant digits for each error condition are listed in Table 2-1, below:

Note: The first error stops the start-up sequence immediately.

B.4.2 Start-Up Error Codes

Table B-1: DASM Controller Start-Up Error Conditions

Value	Significance
1	Failed To Set Timer
2	Failed To Set Baud Rate
3	Failed To Access Receiver Status Reg. For Serial I/O
4	Failed To Start Refresh Clock
5	Failed In Set Up Of Serial I/O
6	Checksum Failed
7	Failed Static RAM Test
11	Failed I/O RAM Test
12	Failed SCSI Interface Test
13	Failed To Start VRTX Operating System
21	Bus Error
22	Address Error
23	Illegal Instruction
24	Undefined MFP (MC68901) Interrupt
25	Zero divide Through Trace Trap Level 5
26	Chk, Trapv, Privilege, Or Trace Interrupt
31	Unknown Interrupt

B.5 LCAM ERROR CODES

The following sequence is used to signal an error during the LCAM portion of the Self-Test.

First, all three LEDs blink together for the duration of a single pass. This requires approximately 10 - 15 seconds. Then the specific error code blinks twice before the next pass begins, with all three LEDs blinking together as before.

The bit pattern of the error codes appears as a series of LEDs being turned on and off. The low order bit is first.

When two LEDs blink together, the bit value is 1.

When only one LED blinks, the bit value is 0.

The Self-Test runs in continuous mode when SW 1 position 0 is "open". A single failure will cause a error code to flash repeatedly on the LEDs. Although, other errors may occur during testing only the first one is displayed forever.

First all LEDs will blink together several times indicates a problem. Next, the error code is displayed serially, twice in the two leftmost LEDs.

The pattern is binary and takes 4 blinks for the complete code. When the two LEDs blink together, the bit value is 1, when only one blinks the bit is 0. The bit order is low to high.

Record the blink pattern in the number of LEDs blinking, 1 or 2.

Record them **RIGHT to LEFT**, to make them easier to interpret.

1 2 1 2 Subtract 1 from each digit to obtain the bit value

0 1 0 1 The value of the code is 5

Error code meanings:

- 1: DMA time-out. Bypass active
- 2: data error. Bypass active
- 3: data error. ROI active
- 4: data error. Bypass, ROI active
- 5: DMA time-out. Swap & bypass active
- 6: data error. Swap & bypass active
- 7: data error. Swap active

8: data error. Swap & ROI active
 9: data error. Swap, ROI active
 14: SRAM LUT error
 No others are defined.

B.6 RUN-TIME ERROR CODES

These codes may be found in the DASM Response Block byte 128, when the error bit (bit 2, where bit 0 is first) is set in Response Block byte 0. Only the first of multiple errors is stored in Response Block byte 128. Reading the Response Block clears the error. Note that the status "invalid command" is returned to the user directly in bit 0 of the SCSI status byte; it is not logged in the Response Block.

Table B-2: Run Time Error Codes

DASM Run-Time Error Codes

Error	Hex	Meaning
— General Error Codes —		
E_PARAM	81	Bad Parameter To Valid Command
E_NO_FF	82	No 0xFF Terminator In Command
E_TMO	83	Timeout
E_PIF_SELF	84	Peripheral Interface Self-Test Error
E_PIFNOTRDY	85	Peripheral Device Not Ready
E_OVRRUN	86	Data Overrun
E_UNDRUN	87	Data Under run
E_COM_LINK	88	Communications Link Error between DASM and PIF
E_DRAM	8A	Error in I/O RAM
E_EPROM	8B	EPROM Checksum Error
E_MFP	8C	Error in the MC68901 I/O and Timer Chip
E_SBIC	8D	Error in Access to SCSI Bus Interface Chip
E_NOTIMER	8E	No Timer Available (All In Use)
E_INTERN	8F	Internal System Software Error
E_INV_MSG	B0	Invalid/unexpected message from device
E_BAD_S	B1	Invalid S-record (from load routine)
E_ODD_ADR	B2	Odd address passed to function
E_BOUNDS	B3	I/O request overlapped buffer boundary

E_DIRECTION	B4	I/O request direction is invalid
E_NOT_NXT	B5	I/O request for data is out of sequence
E_AVAIL_CT	B6	I/O request exceed available data count
E_PARITY	B7	Parity error
E_PIF_REV	B8	No PIF revision level supplied
E_TIMER_ID	B9	Timer id error

E_UNDRUN is a warning, which is available in the trace buffer in the DASM Response Block. During a warning, Byte 128 is not updated.

Note that the following has been removed:

E_SRAM 89 Error in Static RAM

Cont.

Error	Hex	Meaning
M_REJECTED	95	Message Was <REJ>ected By Partner
M_BADREPLY	96	Unexpected Reply Was Received
E_NOTREADY	97	Camera Is Not Ready
E_TIMEOUT	98	EXEC Function Did Not Complete In Time
E_TOOLONG	99	Received Telegram Has Too Many Characters
E_UNEXPECTED	9A	Laser Camera Replied With An Unexpected Telegram
E_BADPARAM	9B	Bad Parameter From Host Found In CMDBLK

— VRTX Error Codes —

E_VRTX	C1-F4	<i>VRTX Error Range</i>
	C1	Task ID Invalid
	C2	No Task Control Block Available
	C3	No Memory Available
	C4	No Memory Block
	C5	Mailbox in Use
	C6	Message of Zero
	C7	Buffer Full
	C8	WAITC is in Progress
	C9	Invalid System Call
	CA	Timeout
	CB	No Message Present
	CC	Queue ID Error

CD	Queue Full
CE	Partition ID Error
CF	Fatal Initialization Error
D0	No Character Present
D1	Invalid Configuration Parameter at Init
D2	Invalid Parameter to PCREATE/PEXTEND
E0	No Component Vector Table
E1	Invalid Component
E2	Invalid Opcode For Component
F0	No Control Block Available
F1	Event Flag Group or Semaphore ID Error
F2	Tasks Pending on Event Flag Group or Semaphore
F3	Event Flag Group or Semaphore is Deleted
F4	Event Flag Group Already Set or Overflow

Appendix C

DASM II LCAM Diagrams

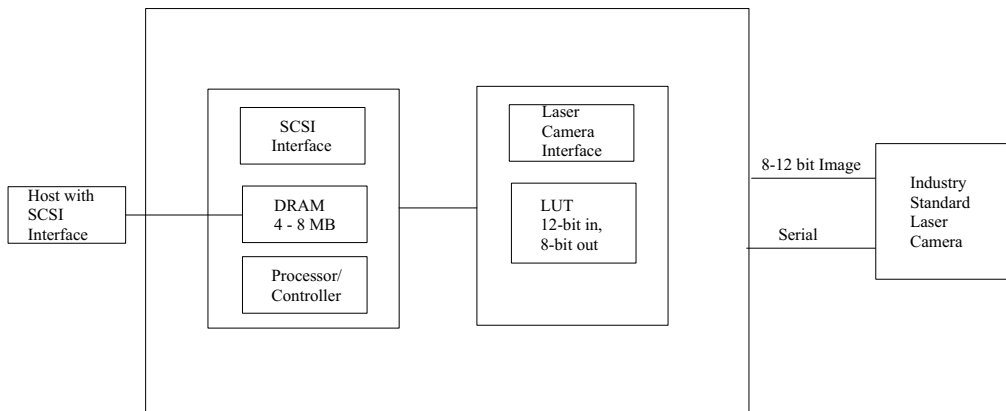
C.1 INTRODUCTION

This section contains the following diagrams for the DASM II LCAM:

- ❑ DASM II LCAM System Block Diagram
- ❑ DASM Controller Block Diagram
- ❑ LCAM Block Diagram

C.2 DASM II LCAM SYSTEM BLOCK DIAGRAM

Figure C-1: DASM II LCAM System Block Diagram



C.3 *DASM CONTROLLER BLOCK DIAGRAM*

The DASM Controller consists of six major blocks, as follows:

- ❑ Processor
- ❑ EPROM
- ❑ Memory Access Controller
- ❑ SCSI Controller
- ❑ Common Data Memory
- ❑ Personality Interface

C.3.1 *Processor (PROC)*

The PROC consists of a MC68001 running at 10MHz, with its own local Static RAM and its own program memory EPROM. The PROC also includes a 68901 multi-function peripheral chip, which handles interrupts and serial communications.

C.3.2 *EPROM*

The EPROM contains the software to operate the DASM II LCAM Controller. The EPROM is connected directly to the 68001 via the CPU Address and Data buses.

C.3.3 *Memory Access Controller (MAC)*

The Common Data Memory is accessed by three different sources: the SCSI DMA, the PIF DMA, and the 68001. The MAC arbitrates the Common Bus. The SCSI DMA and PIF DMA logic are implemented in the MAC. The PROC accesses the DMEM via its own address bus.

The SCSI DMA consists of 24-bit address counters. It transfers data between the SCSI bus and DMEM when allowed by the Common Bus arbitrator. The SCSI DMA releases the Common Bus (COM) if a request is pending and 8-bit data was transferred.

The PIF DMA consists of 24-bit address counters, and 16-bit word counters. It transfers data in burst mode (8 bytes before releasing the COM). The PIF DMA can operate in automatic chaining operation. At the end of a DMA, a new DMA immediately starts without any interaction with the microprocessor. In chaining mode, the new address and word

counter are loaded in temporary registers by the microprocessor during the current DMA.

The COM bus arbitrator gives the highest priority to a refresh cycle. When no refresh request is pending, a "round robin" arbitration scheme is implemented. The DMAs release the COM, only if another request is pending.

C.3.4 SCSI Controller (SCSI)

The SCSI controller is a WD33C93A controller. It includes bi-directional data latches to increase the transfer rate between the SCSI and the DMEM.

C.3.5 Common Data Memory (DMEM)

The Common Memory Data bus is 8-bits wide and may consist of up to 8 Mbytes of Extended Data Out Dynamic Random Access Memory. Memory is organized into two banks, each containing 4 Mbytes. One Kbyte is reserved for the Command Block (Block 0) and the Response Block (Block 1). All other DRAM is available for the user's data.

C.3.6 Personality Interface (PIF)

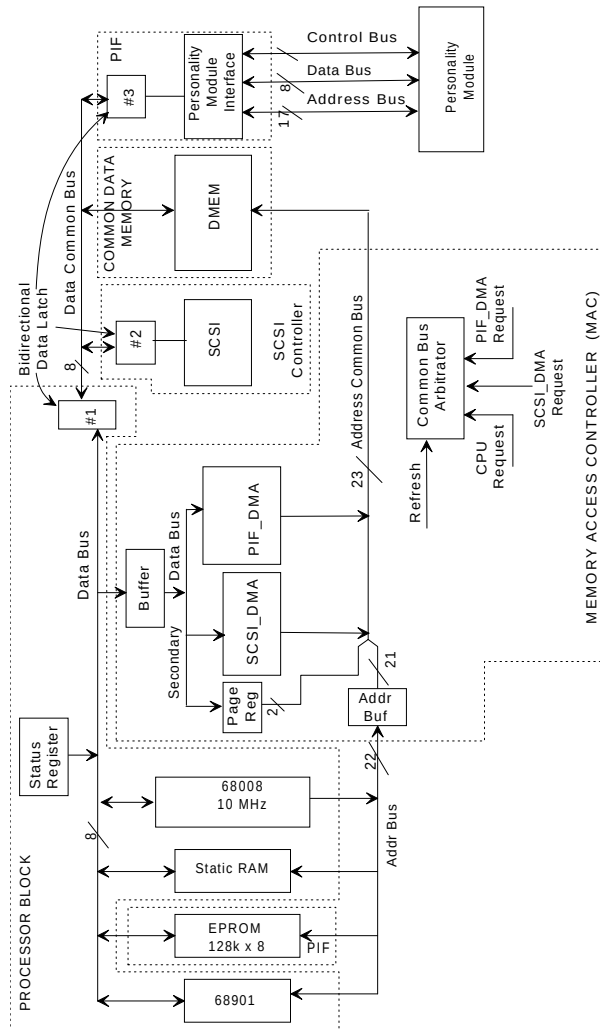
The PIF acts as a slave device, under the control of the 68001 microprocessor, for such operations as writing the Mode Register or reading the Status Register.

During DMA cycles the Memory Access Controller manages PIF activity. Data from the Common Memory is first latched, to permit a higher DMA transfer rate, then enabled on the 8-bit PIF Data bus.

The connections consist of five main groups of signals, as described below.

- ☐ Address lines (17 low addresses)
- ☐ Data Bus (8-bit) from 68001
- ☐ Data Common Bus (8-bit) from (DMEM)
- ☐ DMA handshake signals
- ☐ Serial lines

Figure C-2: DASM Controller Block Diagram



C.4 LCAM BLOCK DIAGRAM

The LCAM transfers data from the DASM to the laser camera. The LCAM is capable of translating 12-bit pixel image data to an 8-bit pixel data through a Look Up Table (LUT). The upper four bits of the input pixel are used to determine whether the data is a Region of Interest (ROI), or an image.

LCAM circuitry consists of four major blocks:

- ❑ Look Up Table (LUT) with Region of Interest (ROI) field detection Byte Swap Module
- ❑ Switch Module
- ❑ Controller
- ❑ Drivers/Receivers

C.4.1 Look Up Table (LUT)

The LUT is loaded by the 68001 with data are received from the host. During data transfers to the laser camera, the twelve bits of data are translated into eight bits by the LUT.

C.4.2 ROI Overlay

ROI overlays, if enabled, are detected during data transfers to the laser camera. If any of the ROI overlay bits are set, the pixel data is converted based upon the MSB of the 8 bit LUT output data. If the Most Significant Bit (MSB) is one, pixel data is forced to all zeros. If the MSB is zero, pixel data is forced to all ones.

C.4.3 Byte Swap Module

The Mode Register (which the 68001 can write) defines the Byte Swap Mode and the MODSEL control lines to the laser camera.

The Byte Swap Module formats the data received from the DASM Controller. It can operate in the following modes:

- ❑ By Pass – This mode bypasses the LUT and sends byte data directly to the laser camera. No ROI Overlay or LUT table translations are done.

- ❑ NO-SWAP – This mode moves word data in from Common Memory to the ROI Overlay and LUT using the following format:
 - byte 0 to MSB
 - byte 1 to Least Significant Bit (LSB)
- ❑ SWAP – This mode moves word data from Common Memory to the ROI Overlay and LUT using the following format:
 - byte 0 to LSB
 - byte 1 to MSB

C.4.4 Switch Module

The Switch Module permits access to the LUT from the 68001 and connects the output of the LUT or the output of the Byte Swap Module to the data lines of the laser camera. In addition, the Switch Mode contains a bypass path that is capable of directing 12 bits to the laser camera. When the Switch Module is connected to the laser camera's data lines, the parity generator creates the parity for DATA11-DATA0. The parity is always even.

C.4.5 Controller

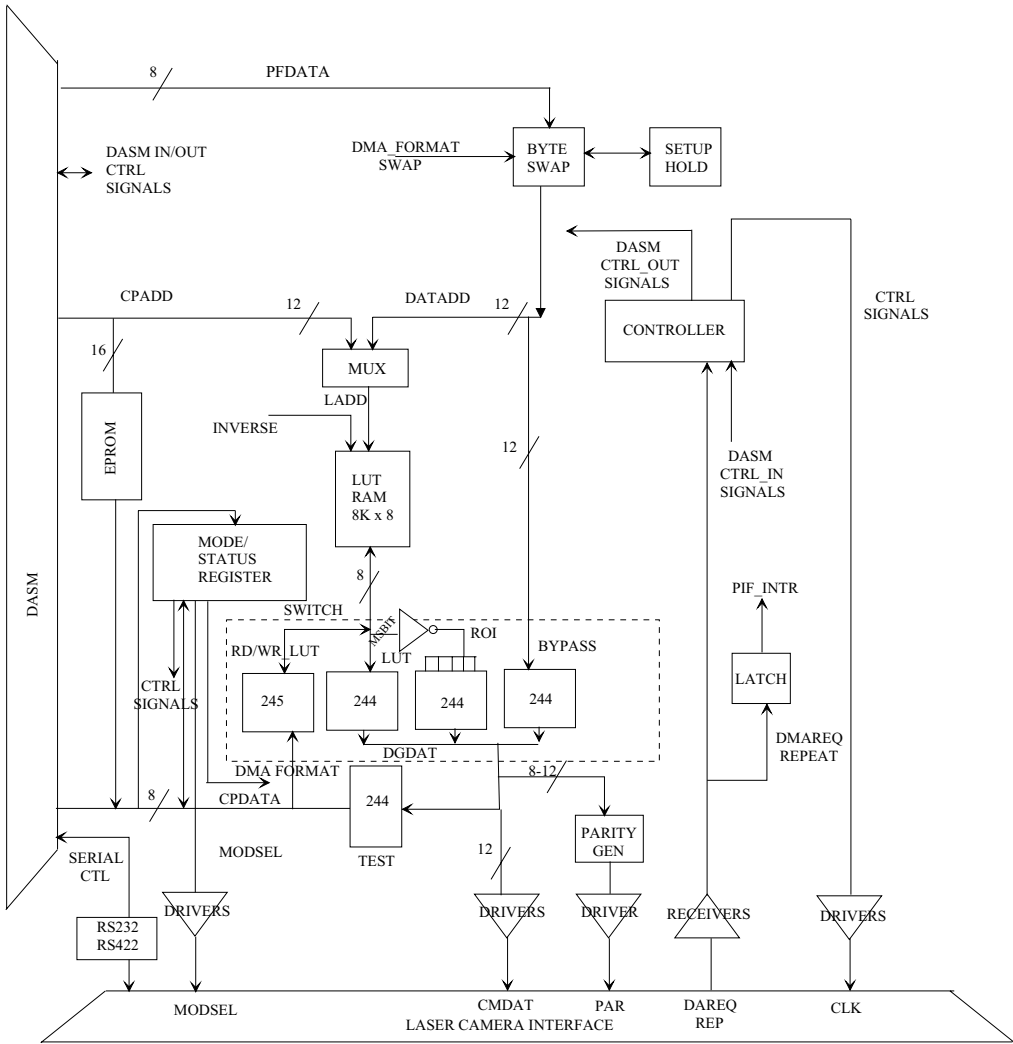
The LCAM Controller handles the handshake protocol between the LCAM Circuitry and the laser camera. It also controls the interface between the DASM Controller board and the LCAM Circuitry.

If the laser camera requires that the last line be repeated due to an error in the transfer (REP signal active), the controller interrupts the 68001. Upon reception of REPEAT Interrupt, the 68001 reloads the last DMA parameters.

C.4.6 Drivers / Receivers

The Image Port drivers and receivers are RS-485 compatible. Pin assignments are described in the DASM II LCAM specifications (Appendix D in this manual).

Figure C-3: LCAM Block Diagram



Appendix D

Specifications

D.1 MECHANICAL

The DASM II LCAM is entirely contained within a single metal enclosure. This enclosure is approximately 10"W x 2.5"H x 10L" and sits on four 1/8" hard rubber feet to prevent scratching of the support surface.

So called "human interfaces" located on rear panel are; a power ON/OFF switch, power entry connector, 25-pin camera control connector, 37-pin image connector, and two 50 pin Centronics type SCSI connectors (SCSI IN and SCSI OUT) with latching cable restraints.

The SCSI ID selection rotary switch, SCSI terminator enable/disable switch and an RS-2323/RS-422 selection switch are accessible on the bottom of the chassis.

The front panel of the DASM II LCAM contains four green LEDs used for status and diagnostics.

D.2 ENVIRONMENTAL

D.2.1 Environmental Conditions

Operating

Operating temperature limits from 0 Centigrade to 40 Centigrade

Operating relative humidity limits from 20% to 93% non-condensing

Operating altitude limits from sea level to 10,000 feet above sea level

Non-Operating

Non-Operating and Storage

Non-Operating temperature limits from -25 Centigrade to 70 Centigrade

Non-Operating relative humidity limits from 20% to 95% non-condensing

D.3 SCSI INTERFACE

SCSI Interface

8-bit SCSI-1, Command Set supported.

SCSI-1 operation is compatible with a SCSI-2 host.

SCSI Transfer Rates

Synchronous Mode: 4.0 Mbytes/sec Max

Asynchronous Mode: 2.5 Mbytes/sec Max

Note that these rates are 'theoretical maximums'. Actual rates will vary. A typical SUN SPARCstation-1 transfers data from DASM memory to SUN memory @ 1.5 Mbytes/sec).

The following SCSI Connector Pin Out table describes the signal associated with each pin in the SCSI cable. All odd pins except pin 25 must be connected to ground. Pin 25 must be left open. Some earlier SCSI devices also may connect this pin to ground. Also note that a minus sign (-) next to the signal indicates active low.

Table D-1: SCSI Connector Pin Out

Signal	Pin Number	Signal	Pin Number
-DB(0)	2	GROUND	28
-DB(1)	4	GROUND	30
-DB(2)	6	-ATN	32
-DB(3)	8	GROUND	34
-DB(4)	10	-BSY	36
-DB(5)	12	-ACK	38
-DB(6)	14	-RST	40
-DB(7)	16	-MSG	42
-DB(P)	18	-SEL	44
GROUND	20	-C/D	46
GROUND	22	-REQ	48
GROUND	24	-I/O	50
TERMPWR	26		

Note: "-" means active low.

D.4 CAMERA INTERFACE CABLE CONNECTOR (J4)*Table D-2: Signal Map from DASM II LCAM to Laser Camera*

Signal Name	LCAM J4 Connector	37-pin Image Connector
DATA9(+)	pin 1	pin 1* on 37-pin connector
DATA9(-)	pin 2	pin 20 on 37-pin connector
DATA8(+)	pin 3	pin 2 on 37-pin connector
DATA8(-)	pin 4	pin 21 on 37-pin connector
DGREP (+)	pin 5	pin 3 on 37-pin connector
DGREP (-)	pin 6	pin 22 on 37-pin connector
DGDAREQ (+)	pin 7	pin 4 on 37-pin connector
DGDAREQ (-)	pin 8	pin 23 on 37-pin connector
DGMODSEL (+)	pin 9	pin 5 on 37-pin connector
DGMODSEL (-)	pin 10	pin 24 on 37-pin connector
DGCLK (+)	pin 11	pin 6 on 37-pin connector
DGCLK (-)	pin 12	pin 25 on 37-pin connector
DGPARG (+)	pin 13	pin 7 on 37-pin connector
DGPARG (-)	pin 14	pin 26 on 37-pin connector
DATA7 (+)	pin 15	pin 8 on 37-pin connector
DATA7 (-)	pin 16	pin 27 on 37-pin connector
DATA6 (+)	pin 17	pin 9 on 37-pin connector
DATA6 (-)	pin 18	pin 28 on 37-pin connector
DATA5 (+)	pin 19	pin 10 on 37-pin connector
DATA5 (-)	pin 20	pin 29 on 37-pin connector
DATA4 (+)	pin 21	pin 11 on 37-pin connector
DATA4 (-)	pin 22	pin 30 on 37-pin connector
DATA3 (+)	pin 23	pin 12 on 37-pin connector
DATA3 (-)	pin 24	pin 31 on 37-pin connector
DATA2 (+)	pin 25	pin 13 on 37-pin connector
DATA2 (-)	pin 26	pin 32 on 37-pin connector
DATA1 (+)	pin 27	pin 14 on 37-pin connector
DATA1 (-)	pin 28	pin 33 on 37-pin connector
DATA0 (+)	pin 29	pin 15 on 37-pin connector
DATA0 (-)	pin 30	pin 34 on 37-pin connector
DATA10(+)	pin 31	pin 16 on 37-pin connector
DATA10(-)	pin 32	pin 35 on 37-pin connector
DATA11(+)	pin 33	pin 17 on 37-pin connector
DATA11(-)	pin 34	pin 36 on 37-pin connector
N.C.	pin 35	pin 18 on 37-pin connector
N.C.	pin 36	pin 37 on 37-pin connector

Signal Name	LCAM J4 Connector	
N.C.	pin 37	
N.C.	pin 38	
N.C.	pin 39	
N.C.	pin 40	
SEND/RS422TX(+)	pin 41	RS232/RS422 (+diff) Output
RTS/RS422TX(-)	pin 42	RS422 (-diff) Output
REC/RS422RX(+)	pin 43	RS232/ RS422 (+diff) Input
CTS/RS422RX(-)	pin 44	RS422 (-diff) Input
RTS/RS422TX(-)	pin 45	RS232 Request To Send
Signal Return	pin 46	DC (logic) Ground
CTS/RS422RX(-)	pin 47	RS232 Clear To Send
Signal Return	pin 48	DC (logic) Ground
N.C.	pin 49	
N.C.	pin 50	

D.5 SERIAL LINE INTERFACE

EIA Standard RS-232 and RS-422A (RS-485 compatible).

Signal Name	LCAM J3 26-Pin	25-Pin Serial Connector
SEND/RS422TX(+)	pin 3	pin 2 RS232 Output
REC/RS422RX(+)	pin 5	pin 3 RS232 Input
RTS/RS422TX(-)	pin 7	pin 4 RS232 Req. To Send
CTS/RS422RX(-)	pin 9	pin 5 RS232 Clear To Send
Signal Return	pin 13	pin 7 DC (logic) Ground
REC/RS422RX(+)	pin 15	pin 8 RS422 (+ diff) Input
CTS/RS422RX(-)	pin 16	pin 21 RS422 (- diff) Input
SEND/RS422TX(+)	pin 17	pin 9 RS422 (+diff)
RTS/RS422TX(-)	pin 18	pin 22 RS422 (- diff)

Note: Because serial signals are multiplexed, cables should only contain wires necessary to implement either RS232 or RS422.

Communications Parameters:

These parameters are software selectable. Default values are listed at the right:

Baud Rate:	standard rates up to 19.2 Kbaud	1200 baud
Word Length:	5,6,7,8 data bits	8 bit
Start/Stop Bits:	0/0, 1/1, 1/1.5, ½	1/1
Parity:	Even, Odd, No Parity	Even
Format:	Synchronous or Asynchronous	Asynchronous

D.6 INPUT

AC Power Requirements

Input range: 100-240VAC
Frequency: 50-60Hz
Current: 0.6A

AC Power Input Connector

The DASM II LCAM provides an industry-standard 3-pin power entry IEC connector for connection to the AC Line.

DC Power Requirements

5.0VDC @ 1.2 Amps Typical, 1.5 Amps Maximum

Combined noise and ripple must be less than 1% of the supplied DC Voltage.

DC Power Mating Connector

Housing part number AMP Incorporated 1-480424-0

Pin part number AMP Incorporated 60619-1

Pin assignment:

J2-4	5.0VDC
J2-3	Return
J2-2	Return
J2-1	Not Connected

SCSI Bus

SCSI-1 single-ended.

Serial Port

RS-422A/RS-485

D.7 FUNCTIONAL GROUNDING

Safety Ground

The DASM II LCAM is provided with a 3-pin AC Power Cord.

This power cord provides the Safety Ground between the exposed metal parts of the DASM II LCAM and the local electrical ground.

D.8 CHARACTERISTICS

DASM Data Memory

Size: 4 Mbytes / 8 Mbytes
Usage: First 1 Kbyte reserved for system use. (Block 0 and Block 1).
All other DRAM available for data.

Physical Characteristics

Standalone enclosure with power supply: 10"W x 2.5"H x 10"D

Front Panel: 4 - green status LEDs for DASM Controller

Rear Panel: 1 - 50-pin "Centronics" female connector: SCSI IN
1 - 50-pin "Centronics" female connector: SCSI OUT
1 - VAC Power Entry IEC Connector
1 - Power Switch
1 - 25-pin "D" connector serial port
1 - 37-pin D connector image port

Cables

AC Power Cord: 6 feet long minimum
SCSI Cable
Length: 3 feet
Connectors: 1 – SCSI-1 50-pin, Centronics-type
1 – SCSI-2 50-pin, subminiature-type
Compatible Hosts: SUN, Hewlett-Packard, IBM RS-6000, others.

External SCSI Terminator Plug

Type: SCSI-1 Single Port, Centronics-type
Mode: Single-Ended Passive or Active

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